

Deep Fake Video Detection Using Attention-Based Convolutional Neural Network

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Abstract—The increasing realism of deepfake videos complicates the verification of digital media. This paper describes a deepfake video detection framework that combines a convolutional neural network (CNN) with a Convolutional Block Attention Module (CBAM) to emphasize informative facial features. The proposed pipeline samples video frames, detects and crops faces, normalizes facial images, and classifies them as authentic or manipulated. Grad-CAM is included to visualize image regions that influence predictions. The evaluation protocol is designed to report accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrices; compare the attention model with an equivalent CNN without attention; test cross-dataset generalization from FaceForensics++ to Celeb-DF; and assess performance after compression and resolution reduction. A FastAPI service and web frontend are specified for video submission and result presentation. Numerical experimental results and implementation-specific model settings were not available for this draft, so no performance claims are made.

Index Terms—deepfake detection, attention mechanisms, CBAM, video forensics, media authentication

I. INTRODUCTION

The growing quality and accessibility of deep-learning-based face manipulation have made it increasingly difficult to distinguish authentic videos from manipulated content by visual inspection alone. Deepfakes can be misused for misinformation, impersonation fraud, and non-consensual content, while uncertainty about authenticity can also undermine trust in genuine media. These risks motivate practical detection methods for digital forensics, journalism, cybersecurity, and content moderation.

CNN-based detectors learn visual representations from facial images and video frames. However, subtle manipulation traces may be concentrated around facial regions such as the eyes, mouth, and blending boundaries. A conventional CNN does not explicitly encode which spatial features should receive greater emphasis. In addition, measured performance depends on manipulation types, dataset composition, split design, compression, and evaluation protocol. Strong results on one benchmark alone do not establish generalization to other datasets or real-world video conditions.

This paper presents a framework for deepfake video detection using a CNN augmented with a Convolutional Block Attention Module (CBAM). Its intended pipeline extracts frames from a submitted video, detects and crops faces, normalizes the

resulting images, and classifies facial content. The framework also includes Grad-CAM visualizations and a web application architecture with a FastAPI backend. Its evaluation plan covers in-domain metrics, a no-attention ablation, cross-dataset testing, and degraded-video conditions. The contributions of this work are:

- A video-level detection pipeline design that combines facial preprocessing with CNN feature extraction and CBAM attention.
- An evaluation protocol for independently examining attention, cross-dataset transfer from FaceForensics++ to Celeb-DF, and robustness to compression and reduced resolution.
- An explainability and deployment design that pairs Grad-CAM visualizations with a FastAPI prediction service and a web-based results interface.

These contributions describe the proposed framework and evaluation design. The available project material does not include completed experimental results or enough implementation details to reproduce a specific trained model; accordingly, this paper does not claim measured improvements.

The remainder of this paper is organized as follows. Section II reviews related work. Section III describes the proposed method. Section IV presents the experimental setup, Section V reports results, and Section VI concludes the paper.

II. RELATED WORK

A. Deepfake Datasets

B. Benchmark Datasets

FaceForensics++ provides manipulated facial videos and a benchmark for evaluating forgery-detection methods [1]. Celeb-DF was introduced as a challenging dataset containing higher-quality synthesized videos, supporting evaluation under more realistic manipulation conditions [2]. Dataset choice and partitioning are consequential: shared identities, source videos, or related frames across training and test partitions can inflate estimates of generalization. Cross-dataset evaluation is therefore useful for measuring domain shift, provided label mapping and preprocessing are documented.

C. CNN-Based Detection

CNN-based systems learn visual patterns that may distinguish manipulated from authentic media. Xception uses depth-wise separable convolutions and has been applied to image classification [3]. Residual networks use identity shortcuts to support deeper CNNs [4]. MesoNet was designed as a compact network for facial video forgery detection [5]. These architectures motivate CNN feature extraction for facial forgery analysis. Comparisons remain meaningful only when datasets, splits, preprocessing, and evaluation units are consistently reported.

D. Positioning of This Study

E. Attention and Explainability

The Convolutional Block Attention Module sequentially applies channel and spatial attention to intermediate CNN features, allowing the network to adaptively emphasize informative feature maps and locations [6]. Grad-CAM uses gradients flowing into a convolutional layer to produce a coarse localization map that indicates image regions influential to a class prediction [7]. In a deepfake detection system, these methods can support attention-guided feature learning and visual inspection of predictions, respectively. A heatmap is an explanation of model sensitivity, however, and should not be interpreted as proof that a highlighted region contains a manipulation artifact.

F. Research Gap and Scope

Prior work establishes CNN-based detection and attention modules as relevant tools, while the benchmark datasets enable evaluation across manipulation sources. This project focuses on bringing several evaluation dimensions into one proposed application: a controlled comparison with an otherwise equivalent CNN without CBAM, cross-dataset testing using FaceForensics++ and Celeb-DF, robustness tests under compression and reduced resolution, and Grad-CAM visualization. This combination is the scope of the proposed framework, not a claim that each component is unprecedented. Its empirical value remains to be established through completed, consistently controlled experiments.

III. METHODOLOGY

A. System Overview

Given an input video, the system samples frames, detects and crops faces, normalizes the crops, and applies a CNN with CBAM to obtain frame-level scores. The valid frame scores are aggregated into a video-level prediction. Grad-CAM is generated for selected frames, and the backend returns the prediction, score, and visualization to the frontend. CBAM operates on feature channels and spatial locations within a frame; it is not a temporal-attention module.

B. Video Sampling and Preprocessing

Let a video be represented by the sampled frames $V = \{I_t\}_{t=1}^T$, where T is the number of sampled frames. A face detector $\mathcal{D}$ and alignment/cropping operator $\mathcal{A}$ produce the facial input

$$F_t = \mathcal{A}(\mathcal{D}(I_t)). \quad (1)$$

The implementation must define how it selects a face when multiple detections occur and how it handles frames with no valid face. After resizing, a pixel value $x_{c,u,v} \in [0, 255]$ may be scaled and standardized channel-wise as

$$\tilde{x}_{c,u,v} = \frac{x_{c,u,v}/255 - \mu_c}{\sigma_c}, \quad (2)$$

where μ_c and σ_c are the selected channel mean and standard deviation; setting $\mu_c = 0$ and $\sigma_c = 1$ gives simple scaling to $[0, 1]$. Their actual values and the input resolution must be reported. During training, augmentation is represented by $\tilde{x}' = \mathcal{T}(\tilde{x})$, where $\mathcal{T}$ is sampled from label-preserving transformations such as horizontal flipping, brightness/contrast changes, or compression simulation. Augmentation is not applied to validation or test data, except in explicitly defined robustness experiments.

C. CNN Feature Extraction and CBAM

A convolutional layer maps an input feature tensor X to output feature maps Y ; for output channel o ,

$$Y_{o,i,j} = \sum_{c=1}^C \sum_{m=0}^{k_h-1} \sum_{n=0}^{k_w-1} K_{o,c,m,n} X_{c,i+m,j+n} + b_o, \quad (3)$$

where K and b denote learned kernels and biases. A rectified linear unit applies $\text{ReLU}(z) = \max(0, z)$. The backbone therefore learns hierarchical spatial representations; the exact backbone and CBAM insertion point are implementation choices that must be reported. Xception and ResNet are candidate backbones [3], [4], but the supplied project description does not identify the selected network, classifier head, initialization, input dimensions, or parameter count. The architecture in Fig. 1 is therefore conceptual and must be checked against the final implementation.

For an intermediate feature tensor $F \in \mathbb{R}^{C \times H \times W}$, CBAM first computes channel attention using spatial average and maximum pooling and a shared multilayer perceptron:

$$\begin{aligned} M_c(F) &= \sigma(\text{MLP}(\text{AvgPool}(F)) \\ &\quad + \text{MLP}(\text{MaxPool}(F))) , \\ F' &= M_c(F) \otimes F. \end{aligned} \quad (4)$$

Here σ is the sigmoid function, and $\otimes$ denotes broadcast element-wise multiplication. Channel attention reweights feature channels. Spatial attention then computes a two-dimensional attention map:

$$\begin{aligned} M_s(F') &= \sigma(f^{7 \times 7}[\text{AvgPool}_c(F'); \text{MaxPool}_c(F')]) , \\ F'' &= M_s(F') \otimes F'. \end{aligned} \quad (5)$$

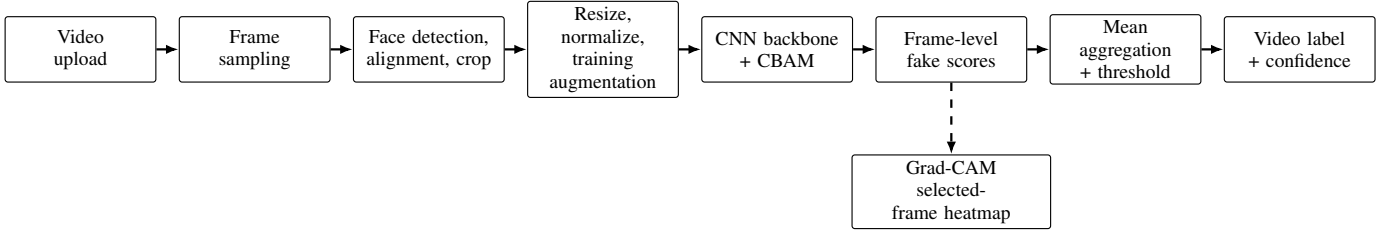


Fig. 1. Proposed video-detection pipeline. CBAM refines spatial feature maps within each sampled frame; Grad-CAM uses the target-class score and convolutional activations to produce a selected-frame explanation.

Here pooling is across channels, $[\cdot]$ denotes concatenation, and $f^{7 \times 7}$ is a convolution with a 7×7 kernel. Thus channel attention addresses which features are informative, while spatial attention addresses where they are informative within a frame. These equations describe CBAM and do not imply that it models relationships across video frames.

D. Training Objective

For binary classification, let $y_i \in \{0, 1\}$ be the label and p_i the predicted probability of the fake class for training example i . The binary cross-entropy objective is

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{N} \sum_{i=1}^N [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)], \quad (6)$$

where N is the number of training examples. Any class weighting or alternative reduction used in training must be stated alongside the loss.

E. Inference

After CBAM refinement, global average pooling produces a feature vector h_t . A binary classifier computes a logit $z_t = \mathbf{w}^\top h_t + b$ and fake-class probability

$$p_t = P(y_t = 1 \mid F_t) = \sigma(z_t) = \frac{1}{1 + e^{-z_t}}. \quad (7)$$

The score is a model probability estimate, not proof of manipulation. Let $\mathcal{T}_v$ be the set of valid sampled frames for video v , with $N_v = |\mathcal{T}_v|$. Mean-probability aggregation gives

$$P_v = \frac{1}{N_v} \sum_{t \in \mathcal{T}_v} p_t, \quad \hat{y}_v = \mathbb{I}[P_v \geq \tau], \quad (8)$$

where τ is a decision threshold selected using the validation set. Mean aggregation is the protocol described here; the final implementation must use and report its actual rule, threshold, frame-sampling policy, and handling of videos with no valid faces.

F. Grad-CAM Explainability

For target class c , let A^k denote the k -th feature map of a selected convolutional layer, y^c the class score, and Z the

number of spatial locations. Grad-CAM computes the feature-map importance weights and heatmap as

$$\alpha_k^c = \frac{1}{Z} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k}, \quad (9)$$

$$L_{\text{Grad-CAM}}^c = \text{ReLU} \left(\sum_k \alpha_k^c A^k \right).$$

The heatmap is resized and overlaid on the corresponding face crop for visualization. It indicates image regions that influence the model score; it is not independent evidence that those regions contain a manipulation artifact [7].

G. Application Architecture

The application is organized into a video-upload frontend, a FastAPI service, video and face preprocessing, model inference, Grad-CAM generation, and a result dashboard. The service packages the predicted label, confidence score, and visualization for the frontend. This architecture describes the intended end-to-end workflow; deployment performance, supported file limits, and operational behavior should be reported after they are verified in the running application.

IV. EXPERIMENTAL SETUP

A. Dataset and Splits

FaceForensics++ is designated as the primary dataset for training and in-domain evaluation, with Celeb-DF reserved for cross-dataset testing. The supplied project description does not identify dataset versions, compression settings, sample counts, manipulation categories included, or actual partition counts. These must be reported from the data used in the final run. Splits should be performed at the source-video level, and, where applicable, identities should not overlap between training and test partitions. All sampled frames from a video must remain in the same partition to prevent leakage. Any filtering, class balancing, or label mapping must also be documented.

B. Implementation Details

The proposed software stack uses Python and PyTorch with torchvision for model and image-processing components, OpenCV for video decoding and frame extraction, and FastAPI for the prediction service. MediaPipe or MTCNN is identified as a possible face detector, but the selected implementation

TABLE I
BINARY CONFUSION MATRIX; THE MANIPULATED CLASS IS POSITIVE.

	Predicted Real	Predicted Fake
Actual Real	TN	FP
Actual Fake	FN	TP

is not specified. The project description also names scikit-learn for metric computation and a web frontend for presenting results. Framework versions, exact hardware, backbone, input resolution, optimizer, learning rate, batch size, epochs, augmentation parameters, random seed, checkpoint-selection rule, and trained model weights were not supplied. These values are necessary for reproducing and interpreting an experimental result.

C. Evaluation Metrics

The evaluation protocol reports accuracy, precision, recall, F1-score, ROC-AUC, and a confusion matrix, with manipulated video defined as the positive class. For a binary classifier, precision and recall are

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}, \quad (10)$$

where TP , FP , and FN denote true positives, false positives, and false negatives, respectively. Accuracy and F1-score are

$$\begin{aligned} \text{Accuracy} &= \frac{TP + TN}{TP + TN + FP + FN}, \\ \text{F1} &= 2 \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}. \end{aligned} \quad (11)$$

The confusion matrix is organized with actual labels by row and predicted labels by column: The receiver operating characteristic (ROC) curve evaluates the true-positive rate (TPR) against the false-positive rate (FPR) over decision thresholds:

$$\begin{aligned} \text{TPR} &= \frac{TP}{TP + FN}, \quad \text{FPR} = \frac{FP}{FP + TN}, \\ \text{AUC} &= \int_0^1 \text{TPR}(u) du. \end{aligned} \quad (12)$$

where u denotes FPR. Metrics should be computed at the video level using the aggregated video prediction, and this unit should be kept consistent across comparisons. ROC-AUC should use continuous prediction scores rather than thresholded labels.

D. Ablation, Generalization, and Robustness

The attention ablation compares the CBAM model with an otherwise equivalent CNN without attention, using identical splits, preprocessing, training budget, and evaluation metrics. Cross-dataset generalization is assessed by training on FaceForensics++ and testing on Celeb-DF without fitting to the target test set. For robustness, let $\mathcal{D}$ denote a specified video degradation, such as compression or spatial downsampling. The transformed test input is $V_d = \mathcal{D}(V)$; performance on V_d is compared with performance on the original V using the

same trained model and decision protocol. Codec, compression quality, target resolution, and any resizing applied before inference must be recorded. No baseline or test result should be presented unless the corresponding experiment has been run under a documented protocol.

V. RESULTS AND DISCUSSION

The supplied project material describes the intended metrics and evaluation scenarios but does not include completed runs, prediction outputs, or verified numerical measurements. Consequently, this section does not report accuracy, precision, recall, F1-score, ROC-AUC, confusion matrices, or comparative gains. Such values cannot be inferred from the proposed architecture or evaluation plan.

The final evaluation should report in-domain results on the held-out FaceForensics++ test partition, the matched no-CBAM ablation, and cross-dataset results on Celeb-DF. Compression and reduced-resolution results should be presented alongside the unmodified test condition, with the degradation settings specified. The discussion should analyze false positives and false negatives, distinguish within-dataset performance from cross-dataset transfer, and limit conclusions to the tested data and conditions.

A. Limitations

The present draft is limited by the absence of a confirmed backbone configuration, training details, data split counts, and experimental measurements. The proposed evaluation also covers only the named datasets and degradation settings; it cannot establish performance across all manipulation methods, populations, or distribution shifts. Compression and resolution changes may remove or alter visual cues, while face-detection errors and frame aggregation can affect video-level decisions. Demographic effects, calibration, and operational latency remain unmeasured in the available material. These are limitations of the evidence currently documented, not findings about the model's measured behavior.

VI. CONCLUSION

This paper describes a proposed deepfake video detection framework that combines facial frame preprocessing, CNN classification, and CBAM attention, with Grad-CAM visualizations and a FastAPI-backed web application architecture. Its evaluation plan includes a controlled no-attention ablation, training on FaceForensics++ with cross-dataset testing on Celeb-DF, and robustness assessment under compression and reduced resolution. These elements address attention validation, domain shift, degraded inputs, and prediction interpretability as explicit evaluation goals.

The available project documentation does not provide a confirmed model configuration or completed experimental results. Therefore, no conclusion about detection accuracy, the benefit of CBAM, cross-dataset generalization, or robustness can yet be drawn. The next step is to finalize and document the implementation and data protocol, run the specified experiments, and report measured results with video-level metrics. Future

extensions may include real-time stream analysis, audio-visual detection, mobile deployment, and training on larger and more diverse datasets.

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